Semantic Understand Instruction Following

Function Description

After the program runs, input a sequence of action commands through the terminal. The large model will plan the action functions corresponding to the action commands and execute all commands in sequence.

Startup

Open the terminal and enter the following command:

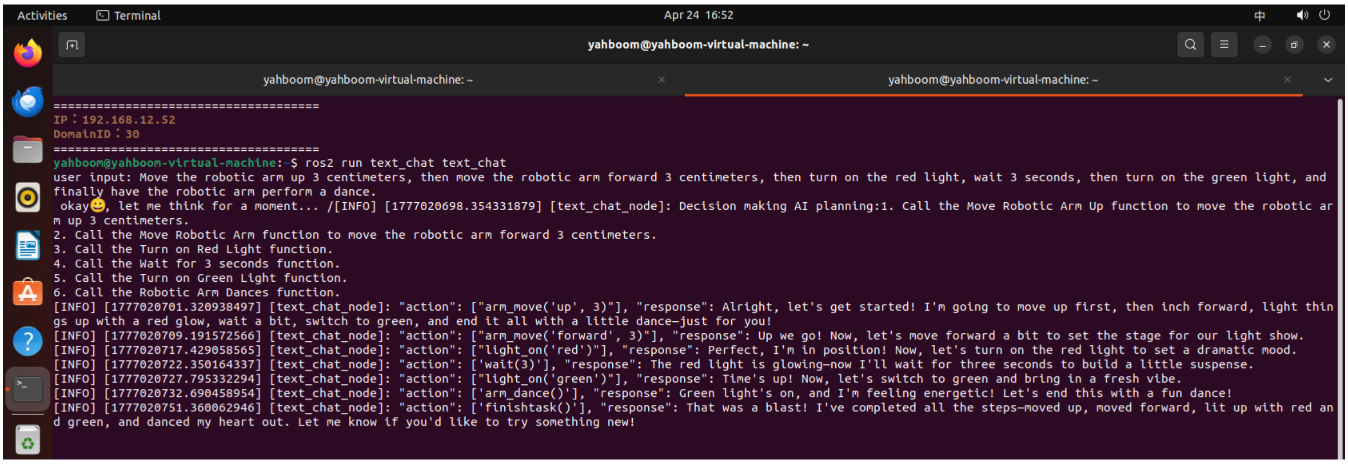
```
ros2 launch largemodel largemodel_control.launch.py text_chat_mode:=True
```

Then open a second terminal and enter the following command:

```
ros2 run text_chat text_chat
```

Then input the action commands you want to execute in the text_chat terminal. You can refer to the following examples:

```
Move the robotic arm up 3 centimeters, then move the robotic arm forward 3 centimeters, then turn on the red light, wait 3 seconds, then turn on the green light, and finally have the robotic arm perform a dance.
```



As shown in the figure above, the large model will plan a series of action command functions, corresponding to the above action commands respectively: **"arm_move('up',3)", "arm_move('forward',3)", "light_on('red')", 'wait(3)', "light_on('green')", 'arm_dance()'**.

The robotic arm end effector will first move forward 3 centimeters, then light up the red LED on the driver board, after waiting for three seconds, the driver board LED will light up green, and finally the robotic arm will execute the dance action.

Core Code Analysis

arm_move Function

`Source code path: LargeModel_ws/src/largemodel/largemodel/action_service.py`

```
#Dir indicates the direction to adjust, Dist indicates the distance to adjust
def arm_move(self,Dir,Dist):
    self.arm_move_flag = True
    cur_joints = [0.0,0.0,0.0,0.0,0.0,0.0]
    #Read the current angle values of six servos
    for i in range(1,7):
        cur_joints[i-1] = Arm.Arm_serial_servo_read(i)
        self.get_logger().info(f"cur_joints[i-1]: {cur_joints[i-1]}")
        if cur_joints[i-1] == None:
            cur_joints[i-1] = 0
            #self.get_logger().info('Servo Reading...')
    self.cur_joints = cur_joints
    Dir = Dir.strip("'\"")  # Remove single and double quotes
    self.Dir = Dir
    Dist = int(Dist)
    self.get_logger().info(f"Dir: {Dir}")
    self.get_logger().info(f"Dist: {Dist}")
    self.get_logger().info(f"cur_joints: {cur_joints}")
    self.cur_joints[0] = float(self.cur_joints[0])
    self.cur_joints[1] = float(self.cur_joints[1])
    self.cur_joints[2] = float(self.cur_joints[2])
    self.cur_joints[3] = float(self.cur_joints[3])
    self.cur_joints[4] = float(self.cur_joints[4])
    self.cur_joints[5] = float(self.cur_joints[5])
    #Get the end effector position of the current robotic arm pose
    self.get_current_end_pos()
    time.sleep(2)
    self.get_logger().info(f"CurEndPos: {self.CurEndPos}")
    #Call the move function to calculate the adjustment
    self.move(Dir,Dist)

def move(self,Dir,Dist):
    while not self.client.wait_for_service(timeout_sec=1.0):
        self.get_logger().info('Service not available, waiting again...')
    request = Kinemarics.Request()
    #Determine direction and calculate the target end position of the robotic arm based on
direction and distance
    if Dir == 'left':
        request.tar_x = self.CurEndPos[0] - Dist*0.01
    elif Dir == 'right':
        request.tar_x = self.CurEndPos[0] + Dist*0.01
    else:
        request.tar_x = self.CurEndPos[0]
    if Dir == 'forward':
        request.tar_y = self.CurEndPos[1]  + Dist*0.01
    elif Dir == 'backwards':
        request.tar_y = self.CurEndPos[1]  - Dist*0.01
```

```
    else:
        request.tar_y = self.CurEndPos[1]
    if Dir == 'up':
        request.tar_z = self.CurEndPos[2] + Dist*0.01
    elif Dir == 'down':
        request.tar_z = self.CurEndPos[2] - Dist*0.01
    else:
        request.tar_z = self.CurEndPos[2]
    request.kin_name = "ik"
    request.roll = self.CurEndPos[3]
    request.pitch = self.CurEndPos[4]
    request.yaw = math.atan(request.tar_x/request.tar_y)
    self.get_logger().info(f"request: {request}")
    future = self.client.call_async(request)

#Call the inverse kinematics service to calculate the target pose of the robotic arm
 def get_ik_respone_callback(self, future):
    try:
        response = future.result()
        joints = [0.0, 0.0, 0.0, 0.0, 0.0,0.0]
        joints[0] = int(response.joint1) #response.joint1
        joints[1] = int(response.joint2)
        joints[2] = int(response.joint3)
        if response.joint4>90:
            joints[3] = 90
        else:
            joints[3] = int(response.joint4)
        joints[4] = 90
        joints[5] = 30
        time.sleep(1.5)
        #If adjusting left/right, only change servo 1 value, keep other values unchanged
        if self.Dir == 'left' or self.Dir == 'right':
            joints[1] = self.cur_joints[1]
            joints[2] = self.cur_joints[2]
            joints[3] = self.cur_joints[3]
         #If adjusting up/down/forward/backward, keep servo 1 value unchanged, change other 
values
        elif  self.Dir == 'down' or self.Dir == 'up' or self.Dir == 'forward' or self.Dir == 
'backwards':
            joints[0] = self.cur_joints[0]
        self.get_logger().info(f"joints: {joints}")
        if self.return_flag == True:
            Arm.Arm_serial_servo_write6(joints[0], joints[1], joints[2], joints[3], 90, 
self.grasp_joint,2000)
            time.sleep(2.0)
            Arm.Arm_serial_servo_write(6, 0, 2000)
            time.sleep(2.0)
        else:
            #Communicate with the bottom control board to control robotic arm movement
            Arm.Arm_serial_servo_write6(joints[0], joints[1], joints[2], joints[3], 90, 
self.grasp_joint,2000)
            time.sleep(2.0)
            for i in range(6):
```

```
                if joints[i] <0:
                    joints[i] = 0
            self.cur_joints = joints
        if self.arm_move_flag == False:
            Arm.Arm_serial_servo_write6(90,120,10,10,90,30,2000)
            time.sleep(2.0)
            if self.back_list == False:
                self.action_status_pub("return_to_orin_done")
            self.return_done = True
        else:
            #time.sleep(2.0)
            self.get_logger().info("Moving.")
            self.get_logger().info(f"self.combination_mode: {self.combination_mode}")
            self.get_logger().info(f"self.interrupt_flag: {self.interrupt_flag}")
            if not self.combination_mode and not self.interrupt_flag:
                self.get_logger().info("Move done.")
                self.action_status_pub("arm_move_done")
    except Exception:
       pass
```

light_on Function

`Source code path: LargeModel_ws/src/largemodel/largemodel/action_service.py`

```
def light_on(self,color):
    color = color.strip("'\"")  # Remove single and double quotes
    self.get_logger().info("Trun on the RGB Light.")
    if color == "red":
        self.get_logger().info("Trun on the Red Light.")
        Arm.Arm_RGB_set(50, 0, 0) #RGB lights up red
    elif color == "green":
        self.get_logger().info("Trun on the Green Light.")
        Arm.Arm_RGB_set(0, 50, 0) #RGB lights up green
    elif color == "blue":
        self.get_logger().info("Trun on the Blue Light.")
        Arm.Arm_RGB_set(0, 0, 50) #RGB lights up blue
    if not self.combination_mode and not self.interrupt_flag:
        self.action_status_pub("light_on_done")
```

wait Function

`Source code path: LargeModel_ws/src/largemodel/largemodel/action_service.py`

```
def wait(self, duration):
    duration = float(duration)
    #Sleep for duration seconds
    time.sleep(duration)
    if not self.combination_mode and not self.interrupt_flag:
        self.action_status_pub("wait_done", duration=duration)
```

arm_dance Function

`Source code path: LargeModel_ws/src/largemodel/largemodel/action_service.py`

```
#Robotic arm dance - directly communicates with the bottom control board to control the robotic arm to move to the specified pose
def arm_dance(self):
    Arm.Arm_serial_servo_write6(90,90,90,90,90,90,1000)
    time.sleep(1.0)
    Arm.Arm_serial_servo_write6(90,60,120,60,90,90,1000)
    time.sleep(1.0)
    Arm.Arm_serial_servo_write6(90,45,135,45,90,90,1000)
    time.sleep(1.0)
    Arm.Arm_serial_servo_write6(90,60,120,60,90,90,1000)
    time.sleep(1.0)
    Arm.Arm_serial_servo_write6(90,90,90,90,90,90,1000)
    time.sleep(1.0)
    Arm.Arm_serial_servo_write6(90,100,80,80,90,90,1000)
    time.sleep(1.0)
    Arm.Arm_serial_servo_write6(90,120,60,60,90,90,1000)
    time.sleep(1.0)
    Arm.Arm_serial_servo_write6(90, 135, 45, 45, 90, 90,1000)
    time.sleep(1.0)
    Arm.Arm_serial_servo_write6(90,90,90,90,90,90,1000)
    time.sleep(1.0)
    Arm.Arm_serial_servo_write6(90, 90, 90, 20, 90, 150,1000)
    time.sleep(1.0)
    Arm.Arm_serial_servo_write6(90, 90, 90, 90, 90, 90,1000)
    time.sleep(1.0)
    Arm.Arm_serial_servo_write6(90, 90, 90, 20, 90, 150,1000)
    time.sleep(1.0)
    Arm.Arm_serial_servo_write6(90, 130, 0, 5, 90, 0,1000)
    if not self.combination_mode and not self.interrupt_flag:
        self.action_status_pub("arm_dance_done")
```